
An Adaptive Market-Making Model for Binary Event Contracts

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Abstract. This report develops a research-grade market-making framework for binary event contracts whose terminal payoff is either zero or one. The market maker trades contracts written on a discrete policy rate (FED) and two correlated private-company valuations (AJR and THR). The framework combines a fitted stochastic factor model, analytic and Monte Carlo probability estimation, inventory-aware reservation prices, competitive RFQ spreads, directional FOK information stress, and exact maximum-loss cash accounting. The design has two ordered objectives: avoid bankruptcy under every admissible execution sequence, then maximise realised PnL and competitive rank. Particular attention is given to the separation between RFQs, where the dealer controls price and displayed size but does not observe the customer's direction, and FOKs, where direction, price and total size are revealed but the dealer can only accept or reject. Version 6 deliberately relaxes ordinary RFQ information charges while stressing an FOK's fair value in the direction implied by surprising price, size, tenor, toxicity and repetition. A state-level caching architecture reduces runtime without changing the economic logic. The report states assumptions, derives the decision rules, maps each rule to implementable Python components, and specifies an out-of-sample validation and model-governance protocol.

Keywords: binary option; event contract; market making; RFQ; fill-or-kill; inventory risk; adverse selection; Monte Carlo; solvency; model governance.

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Executive summary

The system prices an option by its estimated probability of paying one. It then separates the trading decision into a *directional centre* and a *risk width*. The directional centre is the reservation price

$$r = p - \gamma_t C + \theta F,$$

where p is fair value, C is covariance with the existing portfolio, γ_t is dynamic risk aversion, and θF is a learned order-flow adjustment. For an RFQ, the dealer publishes

$$b = \lfloor r - h \rfloor_{0.01}, \quad a = \lceil r + h \rceil_{0.01},$$

where h aggregates base compensation, estimation uncertainty, tenor, live volatility, counterparty toxicity, repeat-request risk and a small tail charge. Displayed quantities are bounded by exact cash capacity and concentration limits.

For a FOK, the customer supplies side, price and total quantity. The dealer accepts only when edge against a directionally stressed reservation price exceeds a residual model hurdle and concentration charge, and when the complete order is solvent under its worst payoff. Large, short-dated, repeated and price-surprising FOKs move the stressed probability further in the customer's direction; ordinary central-probability RFQs may be quoted more competitively. This separation is the central strategic contribution: *compete for ordinary flow without applying the same aggressiveness to orders that reveal unusually strong information*.

Channel	Information observed	Primary control
RFQ	Contract and counterparty; customer side remains hidden	Bid, offer and side-specific displayed quantities
FOK	Contract, customer side, price and total quantity	Binary accept/reject subject to full-order risk

Research standards. The model is deterministic conditional on inputs and seeds, auditable at every decision, explicit about assumptions, and evaluated using walk-forward tests, scenario stress tests, ablations and runtime limits. Bankruptcy prevention is a hard constraint; PnL optimisation is secondary.

1 Problem definition and market microstructure

1.1 Binary contracts

A binary option i has legs $(w_{i\ell}, U_{i\ell})$, strike K_i and expiry T_i . Its observable at expiry is

$$Z_i(T_i) = \sum_{\ell=1}^{L_i} w_{i\ell} U_{i\ell}(T_i),$$

and its cash payoff is

$$X_i = \mathbf{1}_{\{Z_i(T_i) \geq K_i\}} \in \{0, 1\}.$$

Examples include $\mathbf{1}_{\{\text{FED}_{t+1} \geq 1.75\}}$, $\mathbf{1}_{\{\text{THR}_{t+2} \geq 1735\}}$, and $\mathbf{1}_{\{\text{THR}_{t+1} - \text{AJR}_{t+1} \geq 0\}}$. With zero interest and no separate risk-premium adjustment, the model fair value is the physical-model probability

$$p_i(t) = \mathbb{E}_t[X_i] = \mathbb{P}_t(Z_i(T_i) \geq K_i).$$

This is a modelling convention of the simulated market, not a general statement that real-world derivative prices equal physical probabilities.

1.2 RFQ protocol

On an RFQ the exchange requests a two-sided quote

$$Q_i = (b_i, n_i^b, a_i, n_i^a), \quad 0 \leq b_i < a_i \leq 1.$$

The dealer does not know whether the customer will buy or sell. If the customer sells, high bids are competitive; if the customer buys, low offers are competitive. Displayed quantity controls the maximum fill allocated to that side. Consequently, an RFQ decision must price both directions and cannot condition directly on the unrevealed customer side. The exchange may route or split an order across competitive dealers. Adverse selection is weaker than in a revealed-side FOK, but not zero: the best bid or offer can win precisely when another dealer's quote is stale or optimistic.

1.3 FOK protocol

On a fill-or-kill order, the customer provides side $s \in \{\text{BUY}, \text{SELL}\}$, price P , and quantity q . The dealer returns only **True** or **False**. It cannot select a smaller quantity. If several dealers accept, the exchange may split the complete customer order according to its allocation rule; nevertheless each dealer must be capable of carrying all q because the realised allocation is not guaranteed in advance.

1.4 Sign and cash conventions

Let dealer quantity $q_d > 0$ denote a purchase and $q_d < 0$ a sale. Worst-case cash reserved at execution is

$$L(q_d, P) = \begin{cases} q_d P, & q_d > 0, \\ (-q_d)(1 - P), & q_d < 0. \end{cases}$$

A long contract can lose its purchase price; a short contract can lose one minus its sale price. The internal cash ledger deducts this amount immediately. At expiry it releases the payoff for a long or the unused collateral for a short. This convention mirrors the checker and makes solvency a pathwise property rather than an expectation.

1.5 Ordered objective

The strategy solves a lexicographic problem:

1. maintain non-negative cash after every trade and settlement, with a buffer;
2. conditional on survival, maximise realised PnL and the fraction of competitors outperformed.

Expected edge never overrides the hard solvency constraint. A positive-expectation trade can still be rejected if its worst-case loss or concentration is too large.

2 Stochastic factor model and option valuation

2.1 FED dynamics

Let R_t denote the policy-rate state on a grid with step $\delta_R = 0.25$. Conditional on $R_t = r$, the next value is $r + \delta_R$, $r - \delta_R$ (floored at zero), or r . Mean reversion tilts the base transition probabilities:

$$\begin{aligned}\pi_t^\uparrow &= \text{clip}\left(\pi^\uparrow + \kappa(R^* - R_t), 0, 1\right), \\ \pi_t^\downarrow &= \text{clip}\left(\pi^\downarrow - \kappa(R^* - R_t), 0, 1 - \pi_t^\uparrow\right),\end{aligned}$$

with $R^* = 2$. A finite-state recursion prices single-leg FED options exactly and avoids Monte Carlo noise.

2.2 AJR and THR dynamics

Let $V_{j,t}$ be company $j \in \{A, H\}$, where A denotes AJR and H denotes THR. The one-step log return is

$$r_{j,t+1}^{\text{company}} := \log \frac{V_{j,t+1}}{V_{j,t}} = \mu_j + \beta_{j,R} \Delta R_{t+1} + \beta_{j,S} S_{t+1} + \varepsilon_{j,t+1}.$$

The common sector shock and idiosyncratic residuals satisfy

$$S_{t+1} \sim N(0, \sigma_S^2), \quad \varepsilon_{j,t+1} \sim N(0, \sigma_{j,\varepsilon}^2),$$

independently across the two residuals and independently of S . Therefore

$$\text{Cov}(r_{A,t+1}, r_{H,t+1} \mid \Delta R_{t+1}) = \beta_{A,S} \beta_{H,S} \sigma_S^2.$$

The use of log returns preserves positivity before the simulation's cent rounding and turns multiplicative growth into an additive Gaussian factor model. Conditional on a FED path, a single company value is approximately lognormal. The log ratio $\log(V_A/V_H)$ is normal because it is a linear combination of jointly normal log returns; this permits fast analytic pricing of zero-strike AJR–THR comparisons.

2.3 Hybrid pricing engine

The live engine selects the cheapest sufficiently accurate method:

- exact dynamic programming for single-leg FED options;
- conditional lognormal formulae for single AJR or THR legs;
- a conditional normal formula for opposite-sign AJR–THR legs with zero strike;
- deterministic Monte Carlo for unsupported combinations.

The research configuration uses

$$N_{\text{live}} = 1000, \quad N_{\text{theo}} = 5000.$$

The first controls live fallback pricing; the second is reserved for the independent theoretical-pricing interface. Each simulation seed is a deterministic function of parameters, option and state. Repeated calls therefore return identical probabilities, which is essential for attribution and debugging.

2.4 Pricing limitations

The fitted dynamics are deliberately parsimonious. They omit jumps, stochastic volatility, regime changes, endogenous market impact and a separate risk-neutral measure. Daily cent rounding also makes analytic lognormal formulae approximate for some contracts. These limitations enter execution through model-error and volatility buffers rather than being hidden. Production deployment would require independent calibration validation and monitoring for structural breaks.

3 Rolling calibration and uncertainty control

3.1 History window and exponential weighting

The calibration consumes only history actually supplied before trading. It never assumes that 60 or 200 observations exist. The maximum window is $W = 60$ observations and the effective half-life is

$$H = \min\{20, \max(5, (N - 1)/2)\}.$$

With decay $\lambda = \exp(-\log 2/H)$, observation k receives weight proportional to λ^{N-1-k} . The method therefore adapts rapidly under short burn-in while retaining stability when more data are available.

3.2 Rate and company estimation

Weighted up/down indicators estimate the FED transition probabilities. Regressing the difference between up and down indicators on $R^* - R_t$ estimates the mean-reversion coefficient κ . Company log returns are fitted by weighted least squares:

$$r_{j,t+1} = \mu_j + \beta_{j,R} \Delta R_{t+1} + e_{j,t+1}.$$

The weighted residual variances and covariance form

$$\Sigma_e = \begin{pmatrix} \sigma_A^2 & \sigma_{AH} \\ \sigma_{AH} & \sigma_H^2 \end{pmatrix}.$$

A one-factor decomposition chooses sector loadings reproducing σ_{AH} and assigns the remaining non-negative variance to idiosyncratic shocks. Since only the products $\beta_{j,S} \sigma_S$ are identifiable, the implementation normalises $\sigma_S = 1$.

3.3 Shrinkage toward a prior

Short samples are noisy. With prior sample size $N_0 = 30$, the empirical weight is

$$w_N = \frac{N}{N + N_0}.$$

Transition probabilities, mean reversion and residual variances are blended with conservative defaults. Drifts and rate betas are multiplied by w_N . This reduces overfitting without imposing a fictitious 200-day history.

3.4 Estimation-uncertainty spread

For fair value p , the heuristic standard error is

$$\text{SE}(p) = \sqrt{\frac{\max\{p(1-p), 0.01\}}{\max\{N, 25\}}}.$$

The quote adds $a_{\text{model}} \text{SE}(p)$ to its half-spread, with $a_{\text{model}} = 0.15$ in Version 6. At $p = 0.5$ and $N = 25$, this term is 0.015; at $N = 100$, it falls to 0.0075. Thus uncertainty declines smoothly with evidence, but never disappears at extreme probabilities.

3.5 Live volatility and markouts

For each payoff-exposure key, the system records one-step changes in live fair value, excluding terminal zero/one jumps. After at least three observations,

$$\hat{\sigma}_{\text{live}} = \frac{n}{n+8} \sqrt{\frac{1}{n} \sum_{k=1}^n (\Delta p_k)^2},$$

clipped at 0.06. The Version 6 spread contribution is $\min\{0.01, 0.20\hat{\sigma}_{\text{live}}\}$. Execution markouts are also recorded by counterparty and payoff key. A negative direction-adjusted markout means the market moved in the customer's favour and supplies evidence of adverse selection.

3.6 Calibration governance

All fitted parameters, window lengths and priors should be logged at each step. Alerts should cover probabilities at constraints, covariance matrices near singularity, large parameter jumps and systematic forecast miscalibration. Research changes must be tested out of sample; hidden-test scores are not themselves a statistically valid calibration objective.

4 RFQ pricing: reservation price and adaptive spread

4.1 Inventory-aware centre

Let X be the candidate payoff and let the active portfolio be $\Pi = \sum_j q_j X_j$. The relevant covariance is

$$C = \text{Cov}(X, \Pi) = \sum_j q_j \text{Cov}(X, X_j).$$

Dynamic risk aversion is

$$\gamma_t = \text{clip}(\gamma_0[1 + 2D_t + 0.75U_t + 2L_t], 0.01, 0.10),$$

where D_t is realised drawdown, U_t is collateral utilisation and L_t is a recent-negative-PnL fraction. With $\gamma_0 = 0.03$, define

$$s_t^{\text{inv}} = \text{clip}(\gamma_t C, -0.12, 0.12).$$

Positive covariance with a long portfolio lowers the centre, discouraging more buying and encouraging sales; negative covariance recognises hedging benefit.

4.2 Learned flow adjustment

For each payoff key the system fits

$$\Delta p_{t+1} = \alpha + \theta F_t + \eta_{t+1},$$

shrinks θ toward zero and clips it to $[-0.03, 0.03]$. Customer buying creates positive F_t ; selling creates negative F_t ; the signal decays by one half at each step. The price adjustment is $\text{clip}(\theta F_t, -0.02, 0.02)$. The reservation price is consequently

$$r = \text{clip}\left(p - s_t^{\text{inv}} + \text{clip}(\theta F_t, -0.02, 0.02), 0, 1\right).$$

Inventory and flow move both bid and offer in the same direction; they do not merely widen the quote.

4.3 Half-spread decomposition

The symmetric risk width is

$$h = \text{clip}(h_{0,t} + 0.15 \text{SE}(p) + 0.001 \min(\sqrt{T}, 3) + 0.35\tau + \min(0.01, 0.20\hat{\sigma}_{\text{live}}) + h_{\text{repeat}} + h_{\text{tail}}, 0.01, 0.075).$$

The initial base is $h_{0,0} = 0.012$ and its adaptive range is $[0.01, 0.04]$. The repeat and tail charges are

$$h_{\text{repeat}} = \min\{0.010, 0.002(m-1)_+\},$$

$$h_{\text{tail}} = 0.005 \text{clip}\left(\frac{|p-0.5| - 0.30}{0.20}, 0, 1\right).$$

The former binary-payoff term $a_v p(1-p)$ is removed from price width: payoff and tail risk are handled mainly through side-specific quantity and FOK concentration, preventing uncertainty from being charged twice.

Counterparty toxicity blends global and payoff-specific adverse markouts:

$$\tau = \text{clip}(0.25\tau_{\text{global}} + 0.75\tau_{\text{specific}}, 0, 0.10),$$

where both components are shrunk toward zero under limited observations. Toxicity widens both sides because an RFQ does not reveal which side the customer will choose.

4.4 Penny-grid construction

Raw prices are $r - h$ and $r + h$. The dealer rounds outward:

$$b = \max\{0, \min(0.99, \lfloor 100(r-h) \rfloor / 100)\},$$

$$a = \min\{1, \max(0.01, \lceil 100(r+h) \rceil / 100)\}.$$

Outward rounding never silently reduces the intended protection. For $r = 0.5823$ and $h = 0.03$, one obtains $b = 0.55$ and $a = 0.62$. Version 6 caps h at 0.035 only when $0.15 \leq p \leq 0.85$, toxicity is at most 0.015, there is no repeat request, $|s_t^{\text{inv}}| \leq 0.04$, and collateral utilisation is at most 60%. This clean-flow cap directly addresses the one-cent winner-takes-most nature of RFQ competition while retaining wider quotes under observable risk.

5 RFQ size, tail protection and adaptive behaviour

5.1 Worst loss per displayed contract

At bid b , buying one contract has worst loss $\ell_b = b$. At offer a , selling one has worst loss $\ell_a = 1 - a$. Let

$$B_t = f_t^{\text{buffer}} C_0, \quad F_t = (C_t - B_t)_+,$$

where C_0 is initial capital, C_t is internal free cash and $f_t^{\text{buffer}} \geq 0.15$ rises with drawdown, utilisation and recent loss. Cash capacity is $\lfloor F_t/\ell \rfloor$ for positive ℓ .

The target per-fill risk budget is

$$R_t = \min\{0.10C_0, 0.30F_t\},$$

in Version 6. The raw target is $\max\{1, \lfloor R_t/\ell \rfloor\}$. It is multiplied by the toxicity scale $(1 + 8\tau)^{-1}$, the dynamic risk scale, and the portfolio scale

$$k_C = \begin{cases} (1 + 10d_C)^{-1}, & d_C > 0, \\ 1 + 4 \min(-d_C, 0.12), & d_C \leq 0, \end{cases} \quad d_C = d s_t^{\text{inv}},$$

where dealer direction $d = +1$ is a purchase and $d = -1$ a sale. The final displayed size is the minimum of scaled target, cash capacity, the maximum quote size (30), and per-option, underlying and gross-position capacity.

5.2 Risk-reducing priority

A sale against an existing long position, or a purchase against an existing short position, reduces net economic exposure. The size logic therefore permits enough quantity to flatten the position when possible. Nevertheless, the trade still passes cash capacity because the simulated exchange reserves maximum loss trade by trade and does not net collateral automatically.

5.3 Directional tail protection

Model error is particularly consequential when a customer asks the dealer to take the unlikely side of an extreme-probability event. The recommended overlay is

```
if fair_value < 0.20 and current_inventory >= 0:
    bid_quantity = max(1, int(0.50 * bid_quantity))

if fair_value > 0.80 and current_inventory <= 0:
    offer_quantity = max(1, int(0.50 * offer_quantity))
```

At low p , the dangerous side is buying; at high p , the dangerous side is selling. The opposite side is not reduced because it either takes the statistically favourable payoff direction or decreases an existing position.

5.4 Adaptive base spread

After each step the base half-spread changes by

$$\Delta h_{0,t} = 0.04(-\bar{m}_t) + 0.003(\bar{f}_t - 0.30) + 0.005L_t,$$

where $\bar{m}_t$ is average direction-adjusted information markout and $\bar{f}_t$ is average fill rate. Adverse markouts, excessive fills and negative PnL widen the base. Favourable markouts and very low participation narrow it. The update is deliberately slow and clipped to avoid instability.

5.5 RFQ algorithm

1. Price the contract and retrieve portfolio covariance, flow and toxicity.
2. Compute reservation price r and half-spread h .
3. Apply the central-probability competitive cap only under low toxicity and small inventory.
4. Round bid downward and offer upward to whole pennies.
5. Compute side-specific quantities from worst loss, free cash and hard limits.
6. Reduce dangerous tail-side quantity by half; preserve exact-flattening capacity.
7. If one side has zero capacity, quote the zero-worst-loss boundary price with minimum legal size: bid zero or offer one.

The design is intentionally asymmetric: aggressiveness is concentrated where the estimated event probability is central and risk can be diversified; extreme tail exposure is acquired slowly.

6 FOK acceptance and adverse-selection protection

6.1 Directional information stress

Let customer direction be $s = +1$ for BUY and $s = -1$ for SELL. A BUY makes the dealer short and is evidence for a higher payoff probability; a SELL makes the dealer long and is evidence for a lower probability. Price surprise is

$$d_P = \max\{s(P - p), 0\}, \quad u = \max\{0.03, \text{SE}(p) + 0.50\hat{\sigma}_{\text{live}}\}, \quad z = \min\{d_P/u, 3\}/3.$$

Define size, tenor, toxicity and repeat scores

$$q_s = \text{clip}((q - 2)/24, 0, 1), \quad T_s = 1/\sqrt{\max(T, 1)}, \quad \tau_s = \text{clip}(\tau/0.05, 0, 1), \quad m_s = \text{clip}((m - 1)/3, 0, 1).$$

Their interaction produces a capped logit shift

$$I = \min\{2.25, z(0.65 + 1.25q_s + 0.55T_s + 0.75\tau_s + 0.25m_s)\}.$$

The stressed probability and reservation price are

$$p^{\text{adv}} = \text{logistic}(\text{logit}(p) + sI), \quad r^{\text{adv}} = \text{clip}(r + p^{\text{adv}} - p, 0, 1).$$

The diagnostic $1 - e^{-I}$ is an informed-flow score, not a calibrated probability that the contract pays. An extreme customer price is both a signal and compensation; the algorithm does not reject merely because P is far from p .

6.2 Residual edge and concentration

After probability stress, the remaining model hurdle is

$$e_{\text{req}} = \text{clip}(0.008 + 0.12\text{SE}(p) + \min\{0.005, 0.10\hat{\sigma}_{\text{live}}\}, 0.008, 0.06).$$

Size, tenor, toxicity and repetition do not enter again; charging them twice would systematically reject good flow. The separate portfolio charge is

$$e_{\text{conc}} = \min\{0.08, 0.50\gamma_t q p(1 - p)\}.$$

For a customer BUY, stressed edge is $P - r^{\text{adv}}$ and maximum loss is $q(1 - P)$. For a customer SELL, they are $r^{\text{adv}} - P$ and qP .

6.3 Solvency and concentration gates

Let B_t be the dynamic cash buffer. Affordability requires

$$C_t - L_{\text{max}} \geq B_t.$$

For a risk-increasing FOK, the separate loss budget is

$$L_{\text{budget}} = \min\{(C_t - B_t)_+, 0.15C_0\}.$$

A fully risk-reducing order may use up to $0.25C_0$, but still must pass affordability and hard position limits. The final rule is

$$\text{accept} = \text{attractive} \wedge \text{affordable} \wedge \text{within limits} \wedge \text{within loss budget}.$$

6.4 Interpretation of observed behaviour

In deterministic replays, a BUY of 26 at 0.94 with $p = 0.7592$ moves stressed fair to about 0.9466 and is rejected. A BUY of 17 at 0.78 with $p = 0.5823$ moves it to about 0.8397 and is also rejected. Conversely, a BUY of two at 0.99 with $p = 0.7232$ stresses fair to about 0.896 but risks only 0.02 in total and can be accepted. Contract count alone is not risk: payoff-contingent loss and information quality must be considered together.

6.5 FOK implementation skeleton

```
accepted = (
    stressed_edge >= required_edge + concentration_charge
    and cash_balance - maximum_loss >= buffer
    and position_capacity >= order.quantity
    and maximum_loss <= loss_budget
)
return accepted
```


7 Solvency, portfolio controls and runtime architecture

7.1 Exact maximum-loss ledger

Every fill changes internal cash immediately. For dealer purchase quantity $q > 0$,

$$C_{t+} = C_t - qP.$$

For dealer sale quantity $-q < 0$,

$$C_{t+} = C_t - q(1 - P).$$

At expiry with payoff $X \in \{0, 1\}$, a long receives qX and a short recovers $q(1 - X)$. Settlement can only increase reserved cash, whereas execution reduces it. The hard affordability gate is therefore applied before every voluntary FOK and is embedded in RFQ displayed-size capacity.

7.2 Layered risk controls

Control	Research setting	Function
Cash buffer	$\geq 15\%$ of initial capital	Preserves liquidity under consecutive one-sided fills
Per-option cap	200 contracts	Emergency protection against single-event concentration
Underlying cap	300 contracts	Limits common AJR, THR or FED factor exposure
Gross cap	500 contracts	Bounds total operational exposure
Maximum quote size	30 contracts	Limits any individual RFQ allocation
FOK loss budget	15%; 25% if risk-reducing	Rejects excessive all-or-nothing downside
Dynamic risk scale	[0.40, 1]	Reduces displayed size after loss or high utilisation

Contract caps are backstops, not substitutes for price-sensitive cash control. Buying 25 contracts at 0.01 risks 0.25, while buying 25 at 0.99 risks 24.75; a single hard count treats economically different positions as identical.

7.3 Fast state-level covariance

The original implementation repeatedly simulated portfolio covariance after each fill, although the underlying state and fitted parameters had not changed. The fast architecture generates $N_{\text{cov}} = 500$ common market paths once per state and stores each option's sampled binary payoff as an integer bitset. If B_X and B_Y are payoff bitsets, then

$$\widehat{\mathbb{E}}[XY] = \frac{\text{bitcount}(B_X \& B_Y)}{N_{\text{cov}}}.$$

Pair covariance is obtained from this joint mean and the two bit counts, and portfolio covariance is a fast weighted sum. A trade changes q_j but not the simulated joint distribution, so it must not invalidate scenario or fair-value caches. Caches are cleared only when the market state or fitted model changes.

Reducing covariance paths from 1500 to 500 triples the speed of scenario generation but increases Monte Carlo standard error by $\sqrt{3}$. The cache changes the more important dimension: simulation cost is paid once per market state rather than once per order or fill. Live prices are similarly cached by immutable option within the state. The separate theoretical test retains its own deterministic sample count.

7.4 Operational safeguards

Production-style controls should include bounded logging, latency counters, cache-hit rates, exception isolation, invariant checks ($0 \leq b < a \leq 1$ and positive sizes), and a deterministic replay facility. Debug printing must be disabled in scored or live sessions because unbounded output can itself cause operational failure.

8 Evaluation, governance and conclusions

8.1 Validation protocol

A credible evaluation separates parameter fitting from performance assessment. Recommended evidence includes:

1. **Walk-forward tests:** fit only on data available before each session; never use future outcomes to tune a contemporaneous quote.
2. **Competitor diversity:** evaluate against fixed-width, inactive, inventory-aware and situational strategies, with randomised order sequences.
3. **Ablations:** remove covariance, toxicity, flow, adaptive spread and FOK penalties one at a time to identify genuine contribution.
4. **Stress tests:** long one-sided RFQ flow, repeated toxic counterparties, correlated expiries, probabilities near zero/one, small starting capital, and maximum order sizes.
5. **Runtime tests:** record decision latency, simulation calls and total session time separately from PnL.

Primary metrics are bankruptcy frequency, minimum cash, maximum drawdown, realised PnL, rank, fill rate, spread capture, one-step markout, tail loss, exposure by factor, FOK acceptance by size/tenor, and runtime percentiles. Passing a runtime limit only means the strategy completed; it does not create a PnL score. Conversely, a non-bankrupt strategy can receive a low score if it beats no competitor.

8.2 Competition interpretation

A fixed-width dealer can outperform an adaptive model in short sessions because one penny often decides RFQ allocation, its simpler estimate has lower variance, and it accepts more realised payoff risk. Version 6 responds by capping clean central RFQ width and moving much of the tail defence into quantities. It does not imitate fixed width blindly: repeated flow, toxicity, inventory covariance, drawdown and FOK surprise remain active. A realised loss on a sound trade is not by itself a model error; evaluation must separate ex-ante edge, markout and terminal payoff noise.

8.3 Parameter governance

The displayed constants are research starting points, not universal optima. Each change should have an economic hypothesis and an acceptance criterion. For example, tightening the central RFQ spread should increase competitive fills without worsening adverse markouts beyond a pre-specified threshold. Reducing the FOK loss fraction should lower tail loss without eliminating all high-edge small orders. Multiple parameters should not be changed simultaneously when attribution matters.

Model versions should record code hash, parameter set, random-seed convention, training window, test set and exception logs. An independent reviewer should challenge model assumptions, probability calibration, covariance estimation, collateral accounting and the possibility that counterparties condition orders on information absent from the model.

8.4 Conclusion

The proposed market maker is organised around three principles. First, price and risk are distinct: fair value estimates the event probability, the reservation price manages directional exposure, and the spread compensates for uncertainty and information risk. Second, execution protocols require different behaviour: ordinary central-probability RFQs may justify aggressive two-sided competition, while revealed-side FOKs require directional probability stress followed by a residual edge test. Third, solvency is pathwise: every accepted order must be affordable under its worst payoff, irrespective of expected edge. State-level caching makes this discipline computationally feasible without removing the economic controls.

The framework remains a research prototype. Its next development priorities are probability-calibration diagnostics, explicit regime detection, uncertainty intervals for portfolio covariance, and walk-forward optimisation of RFQ aggressiveness subject to a hard bankruptcy constraint.

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